

# Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

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# Unfolding a 4-polytope

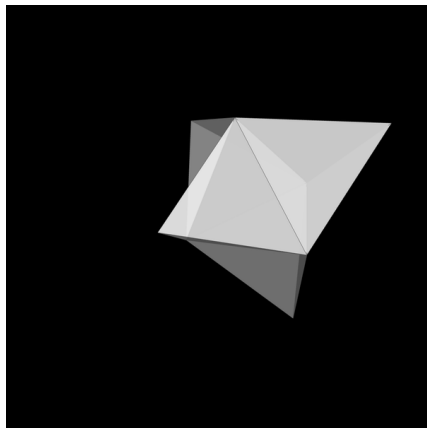
A **net** of a convex polytope: cut it open and lay the facets out flat, without overlap.

In 3D this is Dürer's problem, still open.

In 4D the facets are **3-dimensional cells** and the net lives in  $\mathbb{R}^3$ .

- Turney (1984): all 261 nets of the 8-cell
- Devadoss–Harvey (2023): *every* spanning tree works for the 5-, 8-, 16-cell

**Left open: 24-cell, 120-cell, 600-cell.**



A net of the 5-cell

## Face-rotation BFS

Fix a **root cell**  $r$ . Three steps.

- 1 **Face-up.** Rotate in  $SO(4)$  so the centroid of  $r$  sits on the  $+w$  axis.
- 2 **BFS.** Take a breadth-first spanning tree of the cell-adjacency graph from  $r$ . For each tree edge (parent  $p$ , child  $c$ ), rotate  $c$  about the shared polygon until it lands in  $p$ 's hyperplane.
- 3 **Project.** All cells now share one  $w$ -coordinate (spread  $< 4 \times 10^{-13}$ ). Drop  $w$ .

The result is a net iff no two cells share interior volume.

## The validity test, and a trap

A net is valid if for every pair of cells not joined by a tree edge,

$$\text{Vol}_3(\text{conv}(c_i) \cap \text{conv}(c_j)) \leq 10^{-8}.$$

### Boundary contact is not overlap

Testing with closed-region membership instead counts a shared face, edge or vertex as an intersection. That is common for pointed cells, and it wrongly reports

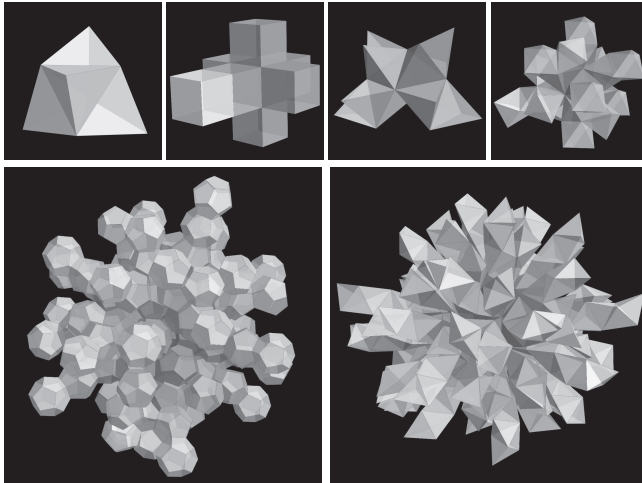
16-cell 0/16, 24-cell 0/24, 600-cell 0/600.

Measuring *volume* ignores contact of measure zero.

# Main result

## Theorem

For every regular convex 4-polytope and **every** root cell, the face-rotation BFS net is valid.



## Contrast: the apple-peel algorithm

A different family of spanning trees — a greedy spiral/zonal ordering (Yoshino–Chaidee).

| Polytope | BFS<br>valid | apple-peel (RZ) |        |
|----------|--------------|-----------------|--------|
|          |              | orderings       | nets   |
| 120-cell | all roots    | 1440/1440       | 0/1440 |
| 600-cell | all roots    | 0/2400          | —      |

The 120-cell has an ordering for every start and **never** a net; the 600-cell has no ordering at all.

Having a valid *ordering* and having a valid *net* are independent properties.

## And more: the tie-breaking question

The abstract left this open:

*“Whether validity extends to all BFS tie-breakings . . . remains open.”*

BFS layers do not depend on the rule, so a BFS tree is just a choice of parent, one layer up, for each cell — and the trees can be counted and enumerated with no rule at all.

| trees per root |                           | checked          |
|----------------|---------------------------|------------------|
| 5-cell         | 1                         | all              |
| 8-cell         | 6                         | all              |
| 16-cell        | 20,736                    | all              |
| 24-cell        | 32,768                    | all              |
| 120-cell       | $\sim 1.8 \times 10^{44}$ | 20 random / root |
| 600-cell       | $\sim 1.8 \times 10^{65}$ | 20 random / root |

**1,132,661 trees, no self-intersection.** For the first four the rule is irrelevant — including the 24-cell, one of the open cases.

## △ **Warning: this does not extend**

Beyond the regular case the rule stops being harmless.

### **Grand antiprism, 320 cells**

rule A    valid from **all 320** roots

rule B    valid from **no** root

Same polytope, same roots, same family of BFS trees. Only the choice of parent differs.

Among the 64 convex uniform 4-polytopes this is not isolated: others swing from 1.8% to 99.8% of their roots.



Grand antiprism, one valid net

**What regularity buys is not that some tree works, but that the choice does not matter.**

## Open problems

- Why? A proof that does not enumerate.
- From BFS trees to **all** spanning trees — still open for the 24-, 120- and 600-cell.
- For the 120- and 600-cell even the BFS family is open:  $10^{44}$  and  $10^{65}$  trees per root put enumeration out of reach, so an argument would have to be structural.
- What separates the uniform polytopes that are insensitive to the rule from those that are not?

Code, data and nets: [github.com/takashi-randomwalker/apple-peel-4d](https://github.com/takashi-randomwalker/apple-peel-4d)

## Backup: “BFS tree” is bigger than what BFS returns

A traversal reaches a tree iff its parent choices admit a total order, i.e. iff the constraint digraph  $p(v) \rightarrow q$  is acyclic.

|         | BFS trees | reachable |       |
|---------|-----------|-----------|-------|
| 8-cell  | 6         | 6         | 100%  |
| 16-cell | 20,736    | 4,896     | 23.6% |
| 24-cell | 32,768    | 2,592     | 7.9%  |

So of the 786,432 trees verified for the 24-cell, only 62,208 are reachable by any traversal. The obstruction is a *cycle* in the constraints, not merely two cells sharing two candidate parents: the 16-cell has no such pair and still leaves three quarters of its trees unreachable.

## Backup: which rules were compared

|                         | queue     | max-index | min-index |
|-------------------------|-----------|-----------|-----------|
| Runcinated 24-cell      | 0/240     | 0/240     | 0/240     |
| Runcitruncated 600-cell | 47/2640   | 2635/2640 | 2627/2640 |
| Runcitruncated 120-cell | 2570/2640 | 2640/2640 | 2151/2640 |
| Grand antiprism         | 320/320   | 320/320   | 0/320     |

“queue” is the textbook BFS loop, neighbours scanned in index order.

“min/max-index” choose, for each cell, the smallest or largest indexed candidate parent.

Uniformly random trees are harsher still: of the 61 uniform polytopes valid from every root under the queue rule, 13 fail within 10 random trees — including all four with 2,640 cells.